

Vibhakar Solanki

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Experience

SDE 2

VideoVerse

- Designed and built **SmartThumbnails** from scratch in **Python/FastAPI** — reverse-engineered selection heuristics from client-published thumbnails into a layered pipeline: **PySceneDetect** scene segmentation, Laplacian sharpness filtering, and **DSPy**-structured Gemini for final selection. Reached **99.2% client acceptance** across the first 4,500 clips, replacing manual curation.
- Built an **SNS-driven** clip enrichment pipeline issuing **DSPy**-structured Gemini calls, so every clip is searchable and tagged at ingest.
- Extended the export pipeline with multi-language, EPG, and AI tagging by routing **SNS events** through the internal AI Gateway, letting broadcast clients localize content without manual work.
- Engineered a hybrid algorithmic + LLM pipeline for sibling-clip metadata back-fill and conflict detection, with batched updates that respect hard SLA windows — cut **SLA breaches by 25%**.
- Unified two fragmented broadcast scheduling sources into a single queryable API; reconciliation logic catches shows split across source APIs that the prior manual-import workflow dropped.
- Owned both the cross-service instrumentation refactor and the collector pipeline for the **Datadog** → **OpenTelemetry** migration, giving the team vendor-neutral structured observability.

Remote

Jan 2026 – present

SDE 1

VideoVerse

- Built an internal LLM evaluation harness from scratch — eval datasets, metric definitions, and **Langfuse** integration — and used it to drive ground-truth analysis that lifted output accuracy **from 68% to 92%**.
- Tuned and applied **DSPy MIPRO/COPRO** and **GEPA** prompt optimizers per service for thumbnail selection and metadata generation, raising task-level accuracy **from 30% to 87%**.
- Built a **FastAPI AI Gateway** fronting **5 backend AI microservices** with service-to-service token auth and retry/backoff baked into both FastAPI middleware and the **HTTPX** transport — **zero dropped jobs** across 50k+ clips and 300+ streams monthly.
- Engineered a **DSPy + Gemini** clip-ratings pipeline running on **500k+ clips/month**, with rate-limit-aware scheduling so peak-load batches still complete before SLA breach.
- Established Langfuse and OTEL alerting workflows to enforce SLA compliance across the AI services.
- Co-led migration of AWS infrastructure to **Terraform**, extending the existing module structure and adding **OTEL** wiring plus a **Cloudflare Tunnel** for secure local access to internal services.

Remote

Nov 2024 – Jan 2026

Software Engineer Intern

VideoVerse

- Originally interned at **LetsDive.io** as a Django backend engineer; continued the role at VideoVerse **following its acquisition of LetsDive.io**.
- Designed an ETL feed from **PostgreSQL and MongoDB** into **BigQuery** with **sub-100ms** end-to-end latency, since live frontend and backend services consume from BigQuery in the request path.
- Rewrote the legacy thumbnail service storage layer from **SQLite to PostgreSQL** with a from-scratch schema (no back-compat), and built the face-embedding ingestion and deployment that drives selection accuracy.

Remote

June 2024 – Nov 2024

Projects

Battlefield Portal Library | [bfportal.gg](#) | <https://gh.bfportal.gg>

- Django/Jinja** CMS, currently migrating templates to **HTMX** for dynamic interactions without a JavaScript framework; **Cloudflare** edge caching, Docker containerization, and GitHub Actions CI/CD.

Jan 2022 – present

Gametools Network | [gametools.network](#) | [github.com/community-network](#)

- Open-source Stats API behind **Envoy** routing into a **Kubernetes** cluster, serving **2M+ daily requests** extended the existing Envoy logic and authored the Kubernetes autoscaling policies for upstream services.

May 2022 – present

Skills

AI / ML: DSPy, MIPRO/COPRO, GEPA, Langfuse, Google Gemini, VertexAI, Prompt Engineering, LLM Evaluation & Optimization

Backend: Python, Go, TypeScript, FastAPI, Django, PostgreSQL, MongoDB, Redis, BigQuery

Infrastructure: AWS (SNS, ECS), GCP (BigQuery, VertexAI), Terraform, Docker, Kubernetes, OpenTelemetry, Grafana, Envoy, Cloudflare, GitHub Actions

Education

MIET

B.Tech in Computer Science Engineering, Minor in Data Science

Meerut, Uttar Pradesh

Jan 2020 – Jan 2024